AMD Ryzen™ AI Max+ 395

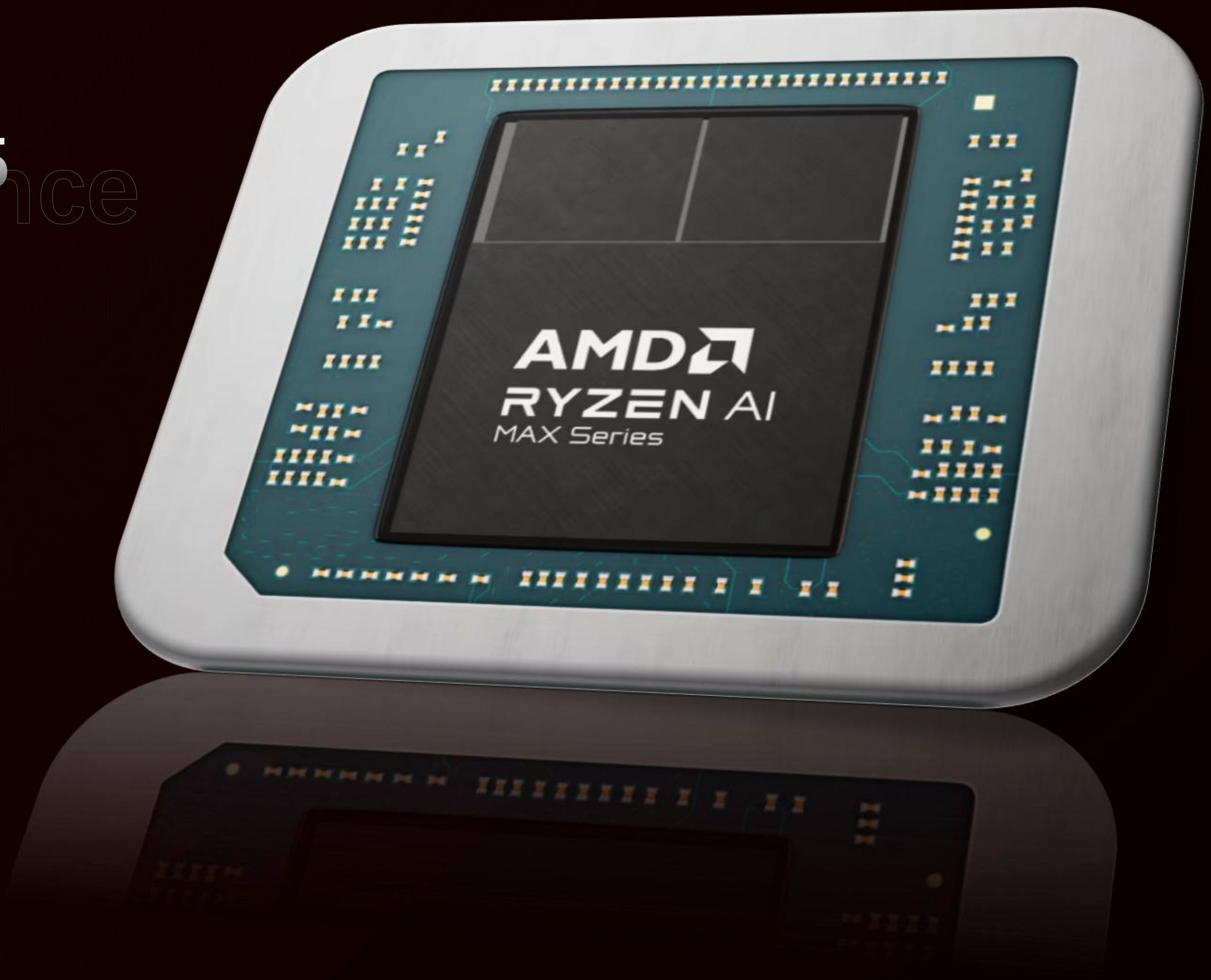
Release on Jan 6th 2025

TSMC N4P	Up to 5.1Ghz	55W TDP	configurable TDP (45-120W)

RDNA 3.5	“Zen5”	XDNA 2

Unified memory architecture

Up to 128GB LPDDR5x-8000

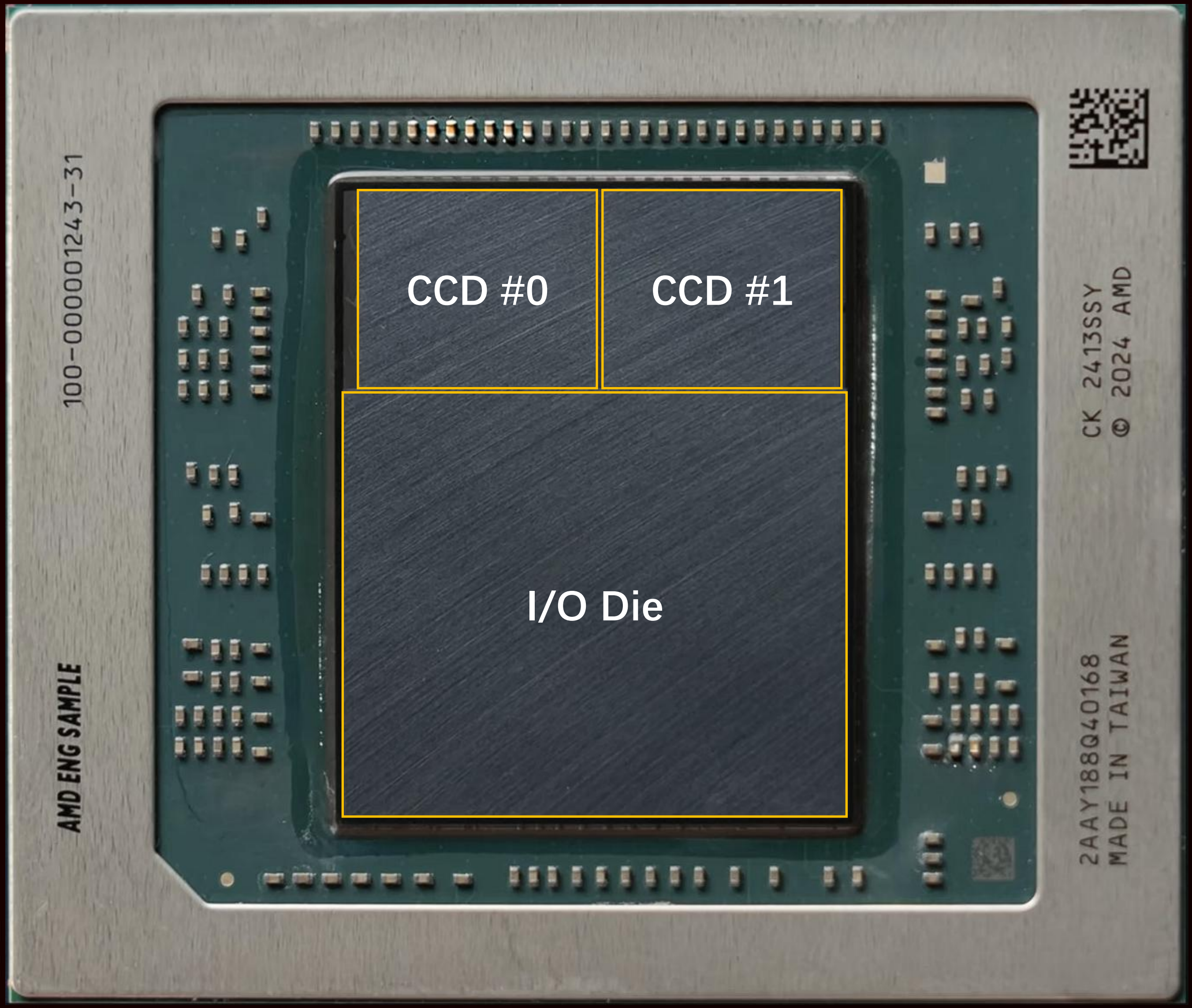


林偉翔01257174 許能翔01257176

2025年计算机结构期末报告

Overview


100-000001243-31
AMD ENG SAMPLE
CK 2413SSY
© 2024 AMD
2AAY188Q40168
MADE IN TAIWAN


CCD #0
CCD #1
I/O Die
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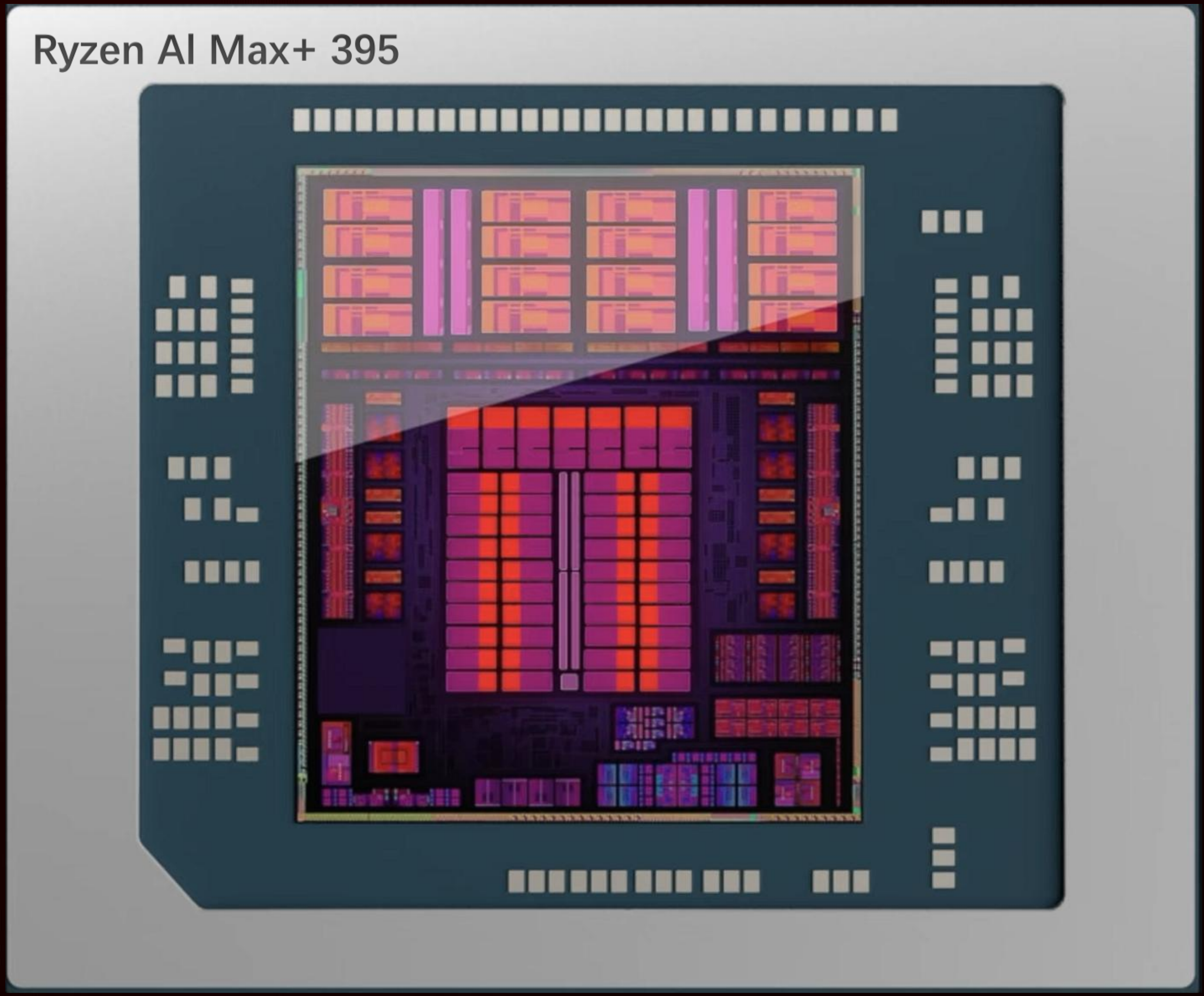
AMD ENG SAMPLE
CK 2413SSY
© 2024 AMD
2AAY188Q40168
MADE IN TAIWAN


Ryzen AI Max+ 395

16-Core CPU (32 Threads)

"Zen5" CPU microarchitecture

Improved branch prediction

8-wide instruction decode

40% larger reorder buffer

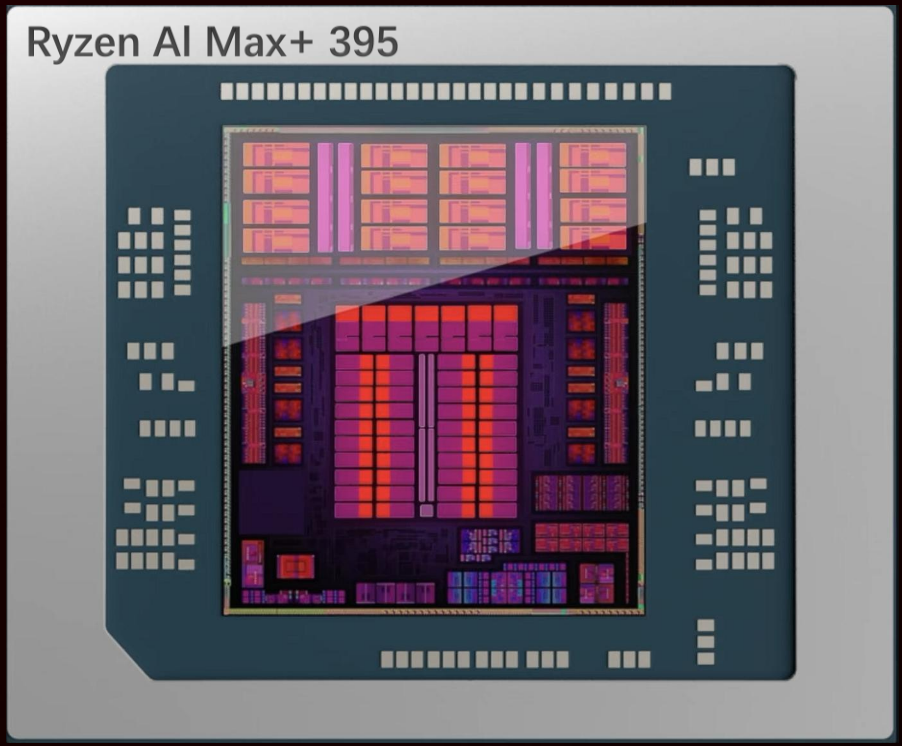
Hierarchy cache memory

16 x 48 KB for L1D

16 x 32 KB for L1I

16 x 1024 KB for L2

2 x 32768 KB for L3



40-CUs GPU

"RDNA 3.5" GPU architecture

2x Texture Sampler Rate

2x Interpolation and Comparison Rates

Improved Memory Management

16-Core CPU (32 Threads)

"Zen5" CPU microarchitecture

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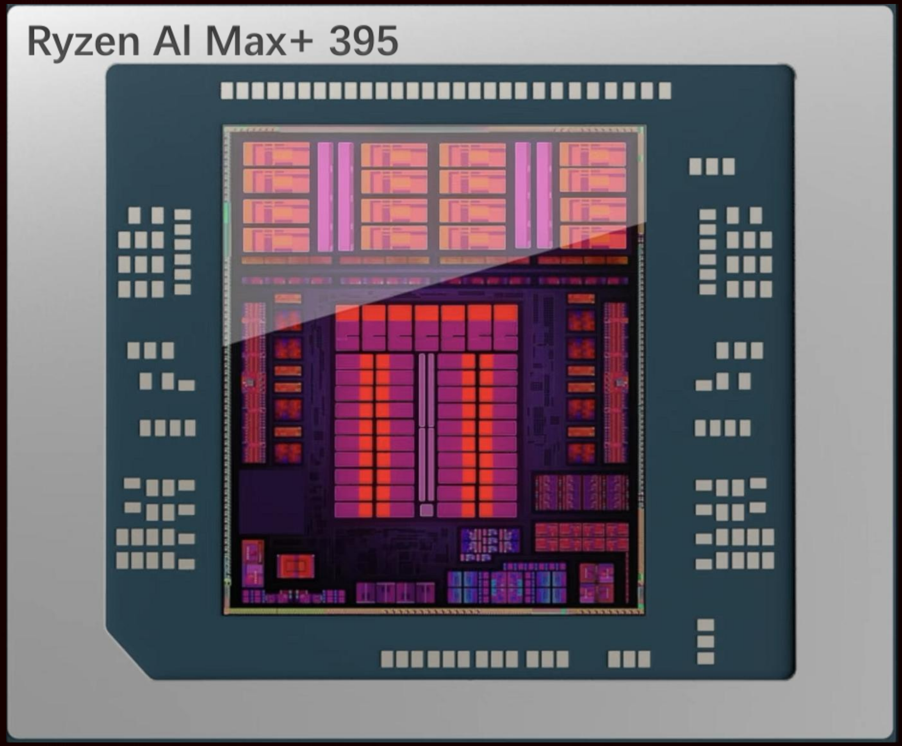


40-CUs GPU

"RDNA 3.5" GPU architecture

2x Texture Sampler Rate

2x Interpolation and Comparison Rates

Improved Memory Management

Next-Gen NPU

"XDNA 2" NPU architecture

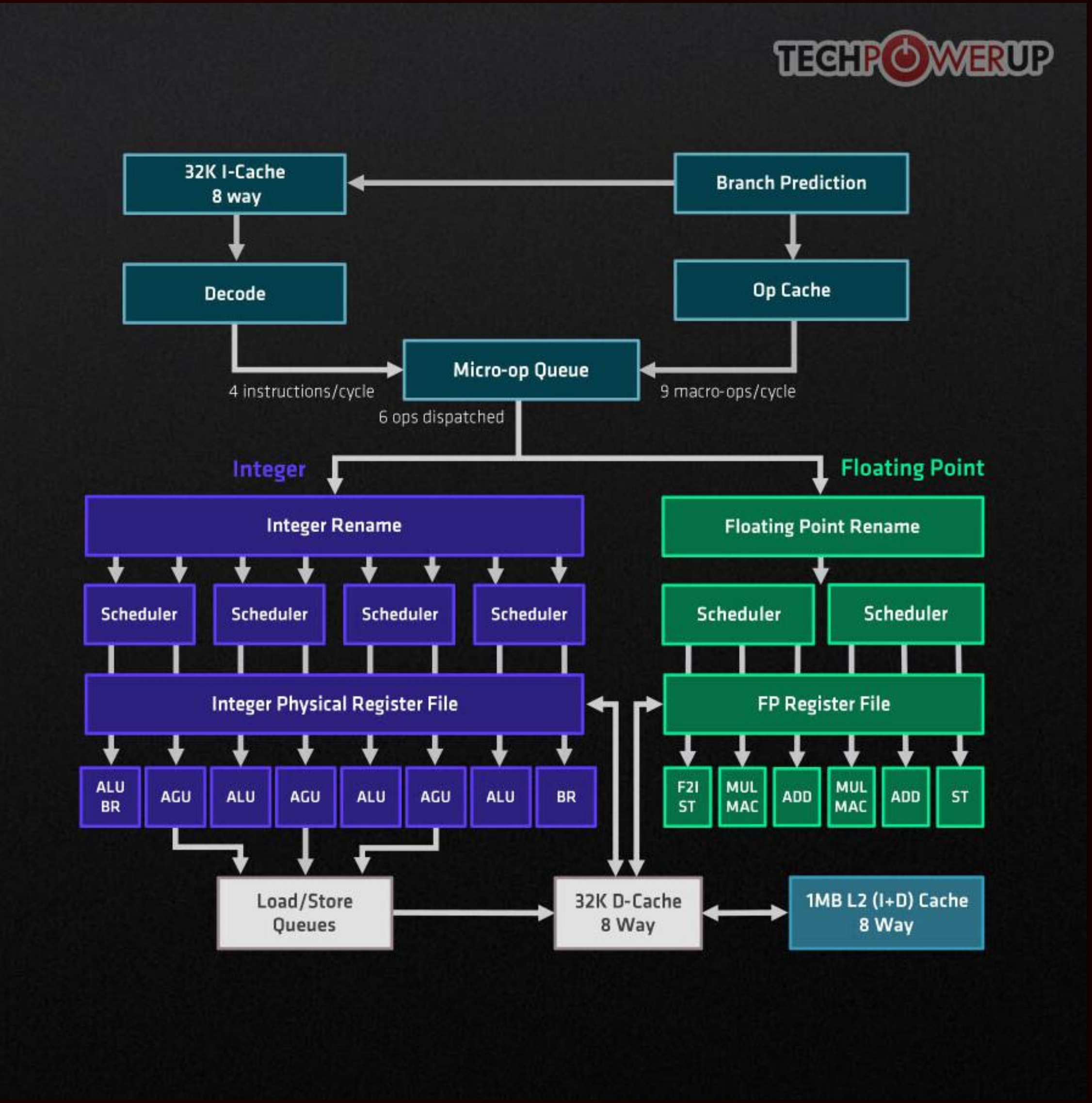
5x compute capacity

2x power efficiency

50 TOPS (Overall 126 TOPS)

About 'Zen5'

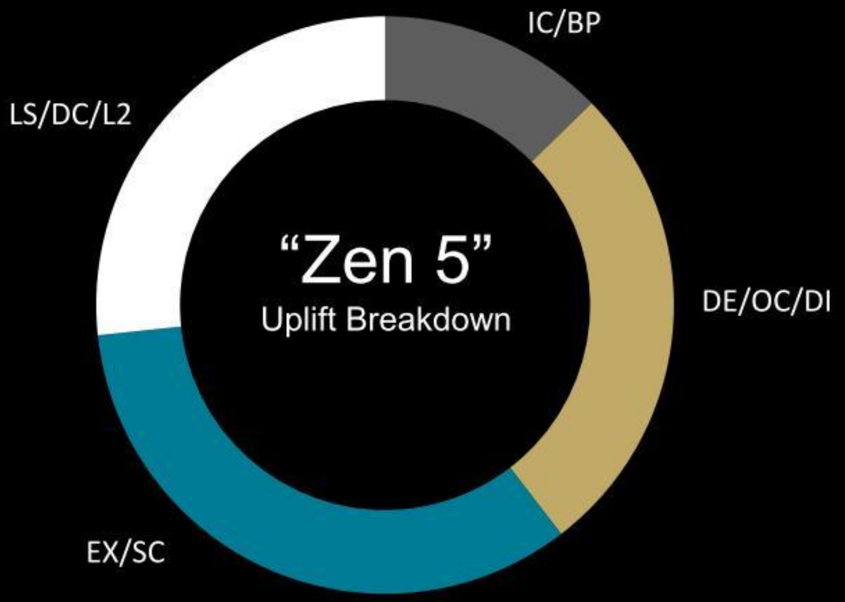
ZEN5


TECHPOWERUP
32K I-Cache
8 way
Branch Prediction
Decode
Op Cache
Micro-op Queue
4 instructions/cycle
9 macro-ops/cycle
6 ops dispatched
Integer
Floating Point
Integer Rename
Floating Point Rename
Scheduler
Scheduler
Scheduler
Scheduler
Scheduler
Scheduler
Integer Physical Register File
FP Register File
ALU BR
AGU
ALU
AGU
ALU
AGU
ALU
BR
F2I ST
MUL MAC
ADD
MUL MAC
ADD
ST
Load/Store Queues
32K D-Cache
8 Way
1MB L2 (I+D) Cache
8 Way


L1I Cache
32KiB 8-way, 32B fetch x2
Branch Prediction
2-taken, 2-ahead, TAGE
Decode
4-wide x2
Op Cache
6Ki inst/fused inst, 6-wide x2
UOPQ
Dispatch, 8-wide
NSQ, 96-entry
Vector Rename, 6-wide
Integer Rename, 8-wide
Sch, 32-entry
Sch, 32-entry
Sch, 32-entry
Scheduler
Scheduler
Vector Registers, 512b, 384-entry
General-Purpose Registers, 64b, 240-entry
FMUL FMA
FADD
FMUL FMA
FADD
StD IntD
StD IntD
ALU Mul
ALU Mul
ALU Mul
ALU BR
ALU BR
ALU BR
AGU
AGU
AGU
AGU
Load/Store Queues
L1D Cache
48KiB 12-way
4 read, 2 write
64B fill, 64B victim
L2 Cache
1MiB 16-way

Key "Zen 5" vs. "Zen 4" Capabilities

Attribute	Zen 4	Zen 5
L1/L2 BTB	1.5K/7K	16K/8K
Return Address Stack	32	52
ITLB L1/L2	64/512	64/2048
Fetched/Decoded Instruction Bytes/cycle	32	64
Op Cache associativity	12-way	16-way
Op Cache bandwidth	9macro-ops	12 inst or fused inst
Dispatch bandwidth (macro-ops/cycle)	6	8
AGU Scheduler	3x24 ALU/AGU	56
ALU Scheduler	1x24 ALU	88
ALU/AGU	4/3	6/4
Int PRF (reg/flag)	224/126	240/192
Vector Reg	192	384
FP Pre-Sched Queue	64	96
FP Scheduler	2x32	3x38
FP Pipes	3	4
Vector Width	256b	256b/512b
ROB/Retire Queue	320	448
LS Mem Pipes support Load/Store	3/1	4/2
DTLB L1/L2	72/3072	96/4096
L1Data Cache	32KB/8-way	48KB/12-way
L2 per core	1MB/8w	1MB/16w
L2 bandwidth	32B/clk	64B/clk

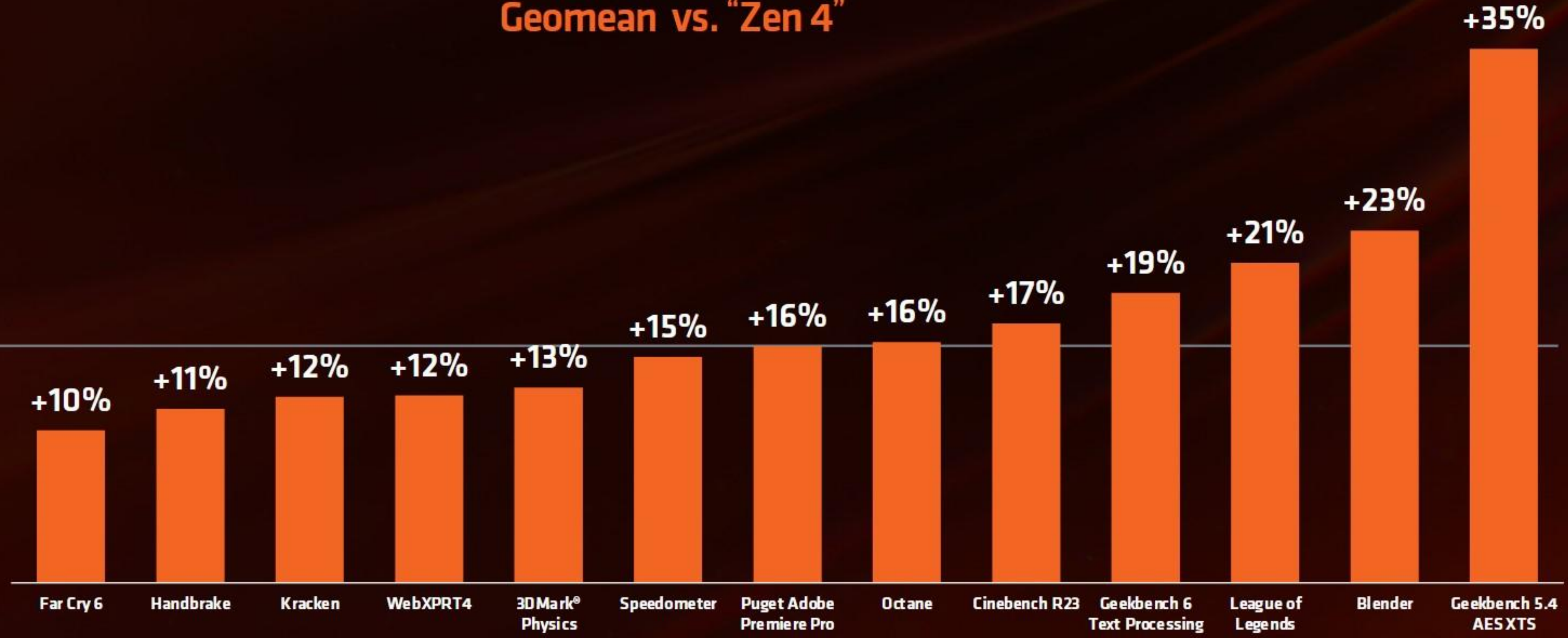


AMD

“Zen 5” IPC Uplift for PCs

Geomean vs. “Zen 4”

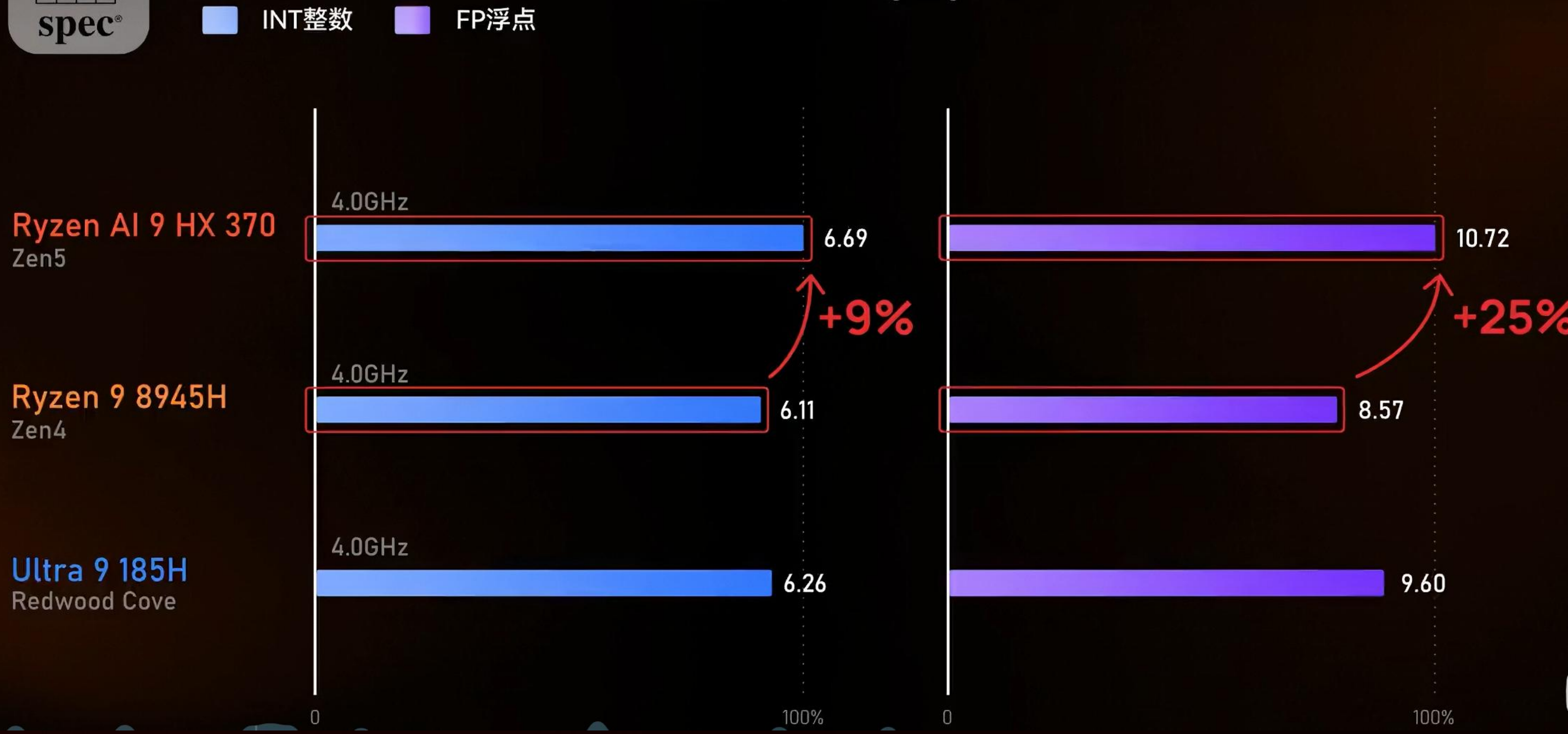
16%
AVERAGE
IPC UPLIFT



*all results are 'up to'. See endnote GNR-03

SPEC 2017 CPU测试

编译环境：Ubuntu 24.04 LTS x86_64, Linux 6.8.0-39-generic
编译器：llvm clang 14.0.6, flang-new 17.0.6
Flags：-g -Ofast



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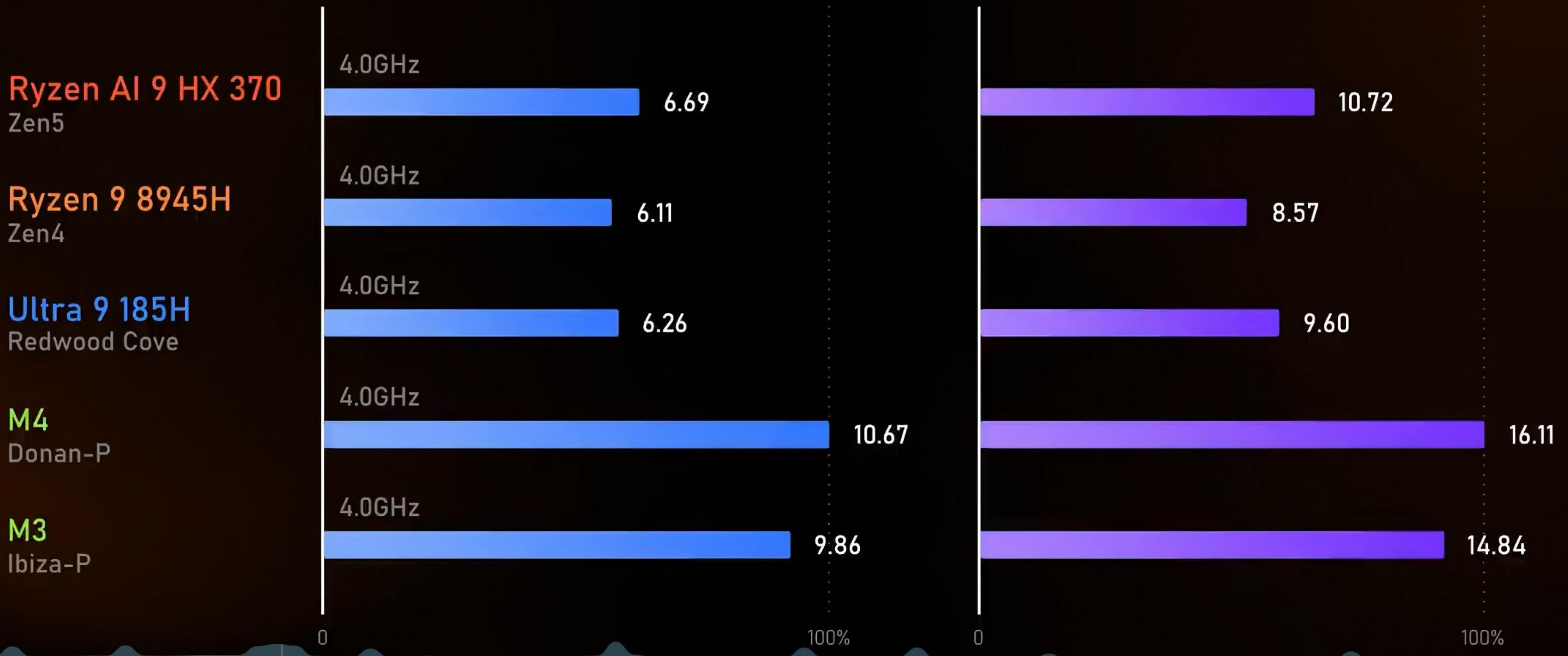
INT整数　FP浮点



RDNA 3.5

RDNA™3: PREMIUM ADVANCED GRAPHICS

INDUSTRY DEFINING CHIPLET ARCHITECTURE

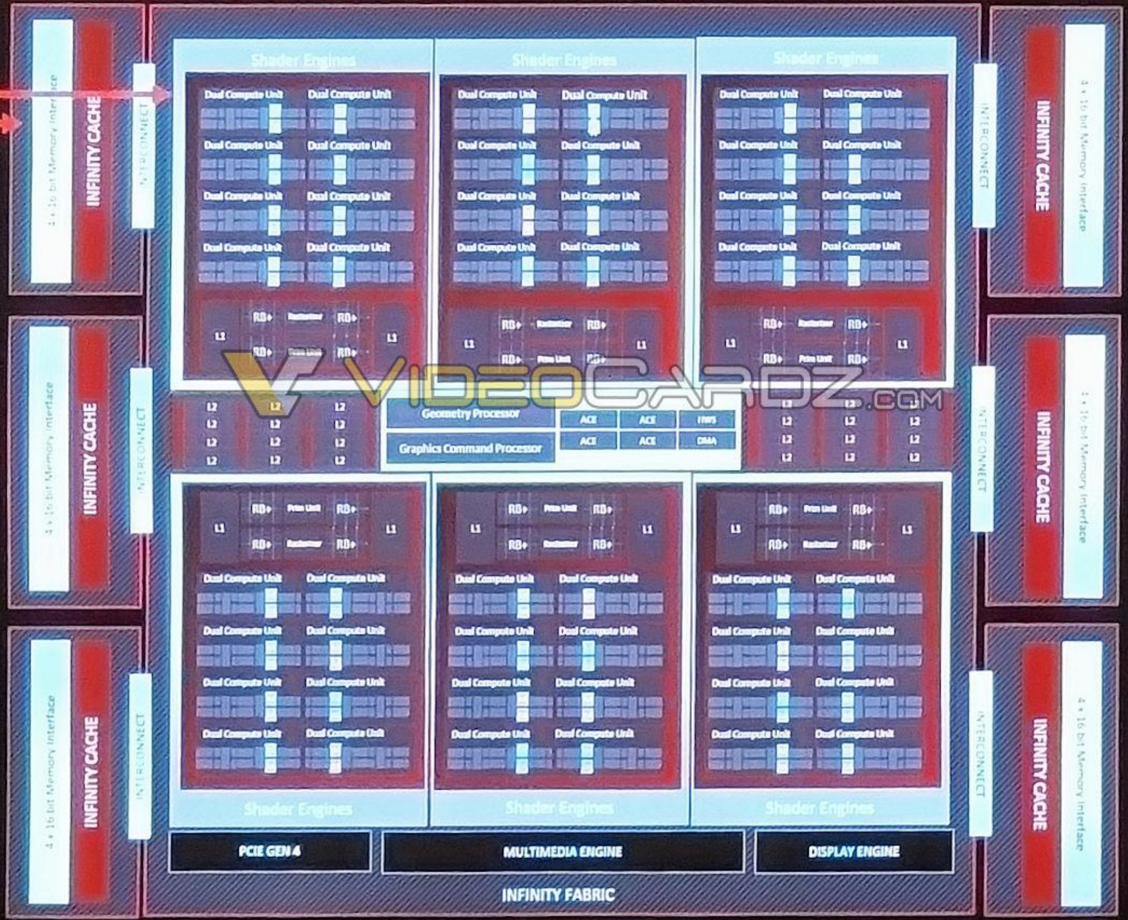
Advanced Chiplet Design
- Disruptive Architecture vs Monolithic
- 5nm high performance Graphics Die
- 6nm Memory Cache Dies (MCD)
- Advanced technology – 1st in gaming

Architected to exceed 3Ghz – Industry 1st
- 61.6 TFLOPs Boost FP32 – ~2.7x increase
- ~1.54x Perf/Watt for a 3rd Generation

New ALUs Instructions and Throughput
- Up to 2x ALU rates plus BF16 support
- New Instructions for effective utilization

Optimized & Balanced Cache System
- 96 MB 2nd Gen Infinity Cache
- 6MB L2 Cache – 50% Increase
- 3MB L1 Cache – 300% Increase
- 3MB L0 Cache - 240% Increase

2nd Generation Ray Tracing
- RT Features for Performance & Efficiency
- Larger Caches for Complex RT Workloads
- Up to 1.8x RT performance @2.505GHz



Flexible CP & Geometry Pipe
- Multi Draw Indirect Accelerator (MDIA)
- 12 Primitive/Clk – 50% Increase
- 2x Hardware Prim/Vert Cull Rates

Advances in Pixel Pipe
- 6 Prims Rasterized/Clk – 50% Increase
- 192 Pixels/Clk – 50% Increase
- Random Order Opaque exports
- Pixel Wait Sync

High Speed GDDR6 Memory
- Up to 384b @ 20 Gbps – 960 GB/s
- Up to 24 GB of GDDR6

AMD Radiance Display™ Engine
- DisplayPort™ 2.1 & HDMI 2.1a
- 12 bit/channel for up to 68 billion colors

New Dual Media Engine
- Simultaneous Encode/Decode (AVC/HEVC)
- 8k60 AV1 Encode/Decode
- AI Enhanced Video Decode

Leading Features
- Full DirectX12 Ultimate
- Fidelity FX Super Resolution
- AMD Advantage Smart Technologies

AMD
together we advance_gaming

SEE ENDNOTES RX-808, RX-809, RX-816, RX-818, RX-821, RX-844

AMD RDNA 3.5

Leveraging the deep collaboration with our mobile partners

Optimized AMD RDNA™ 3 architecture for power-performance efficiency

Optimized for performance/watt

Improve the performance and efficiency of many common graphics shader operations

Optimized for performance/bit

Access memory less often and more efficiently

Designed for better battery life

Advanced power management in the GPU to save active power

AMD RDNA 3.5

Mobile Optimized Features

PERFORMANCE/WATT

2x Texture Sampler Rate

Subset of the most common texture sampling operations are now double rate for common game texture ops

PERFORMANCE/WATT

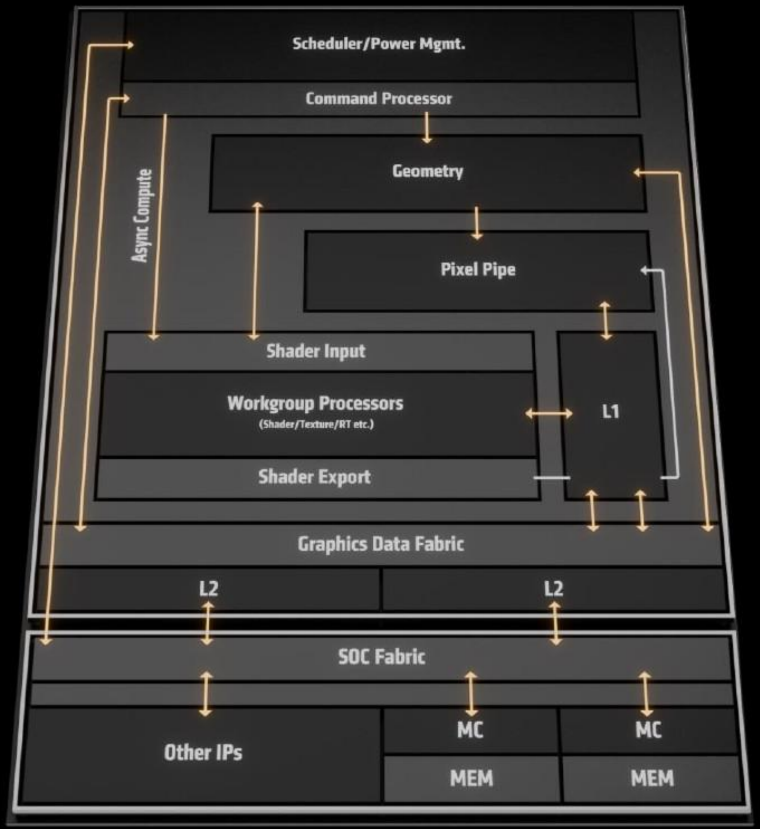
2x Interpolation and Comparison Rates

Most of the rich vector ISA for interpolation and comparison is now double rate for common Ops in shaders

PERFORMANCE/BIT

Improved Memory Management

Improvements to primitive batch processing to reduce memory accesses, better compression techniques, workload reduction and optimized LPDDR5 access



XDNA 2

AMD XDNA™ 2 architecture leadership

AMD XDNA

20
AI Engine Tiles

10
NPU TOPS

Performance uplift

2x MACs per tile

1.6x on-chip memory

Block floating point

Enhanced non-linear support

AMD XDNA 2

32
AI Engine Tiles

50
NPU TOPS

Up to

5x compute capacity

*Compared to Ryzen™ 7040 Series

Up to 8 concurrent spatial streams enabling improved multi-tasking

Up to

2x power efficiency

*Compared to Ryzen™ 7040 Series

Column-based power gating for longer battery life

AMD

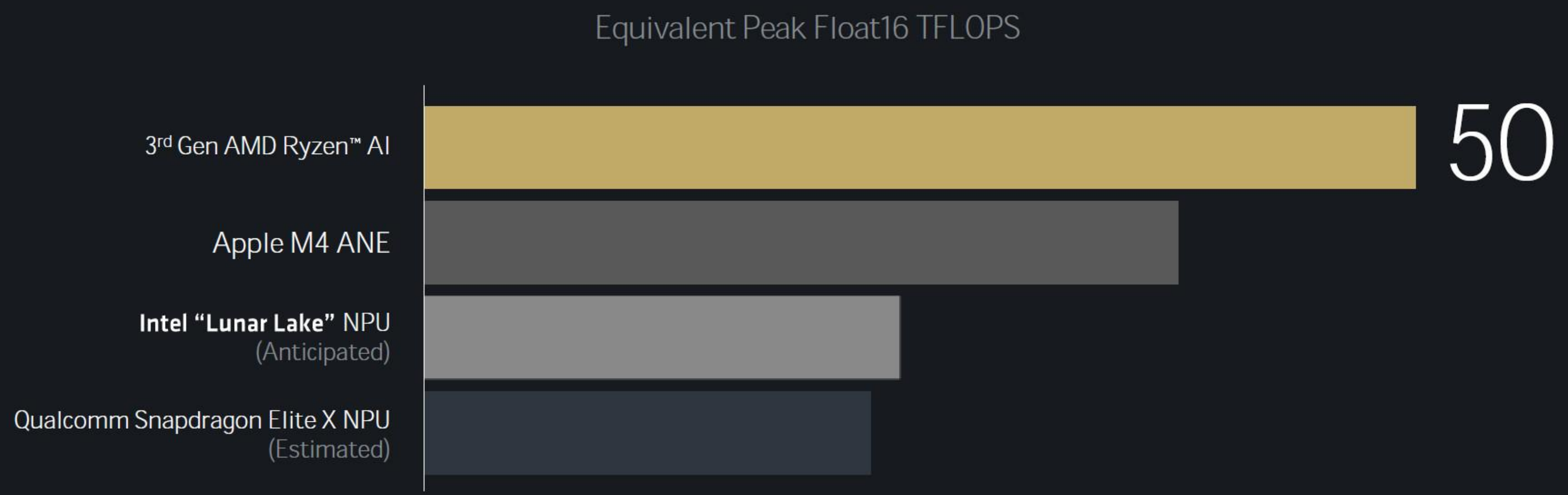
XDNA 2

See endnote STX-13, STX-14

AMD
together we advance_

AMD XDNA 2

World's most powerful NPU for next-gen AI PCs



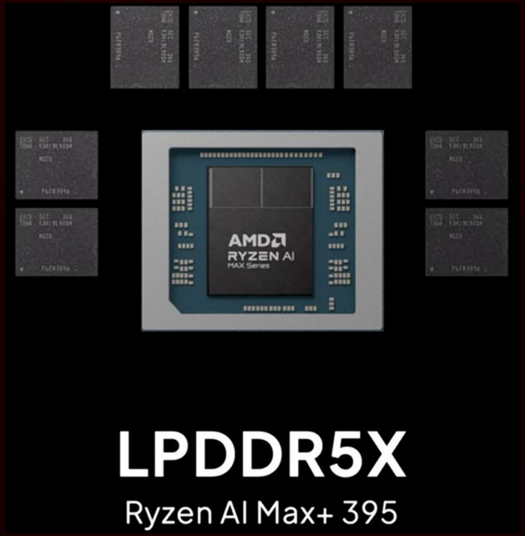
No quantization required. Easy-to-use data types with AMD Ryzen™ AI software stack.

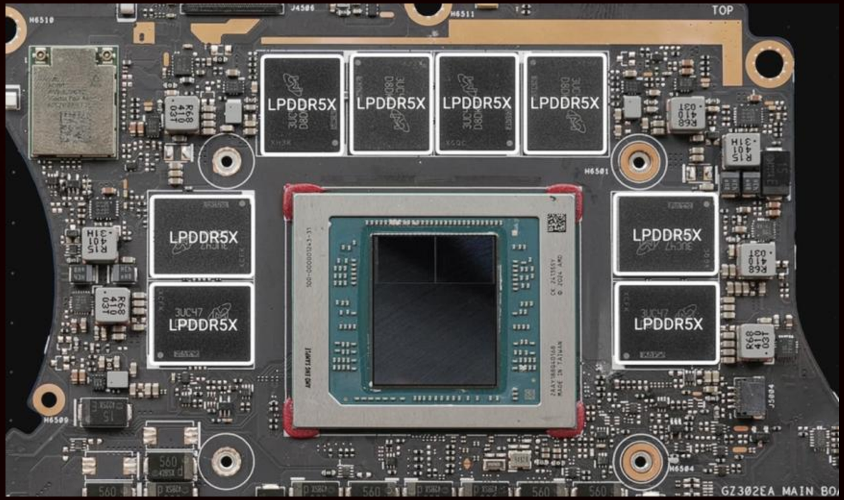
See endnote GD-243, STX-04, STX-44

统一内存架构

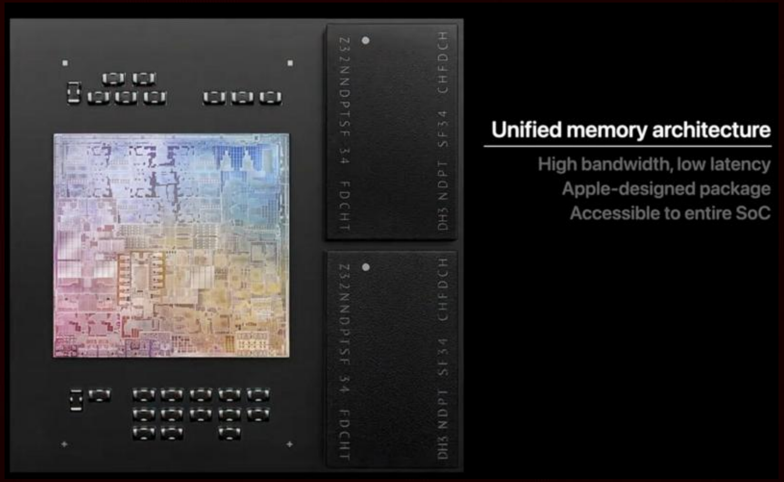
Unified Memory Architecture

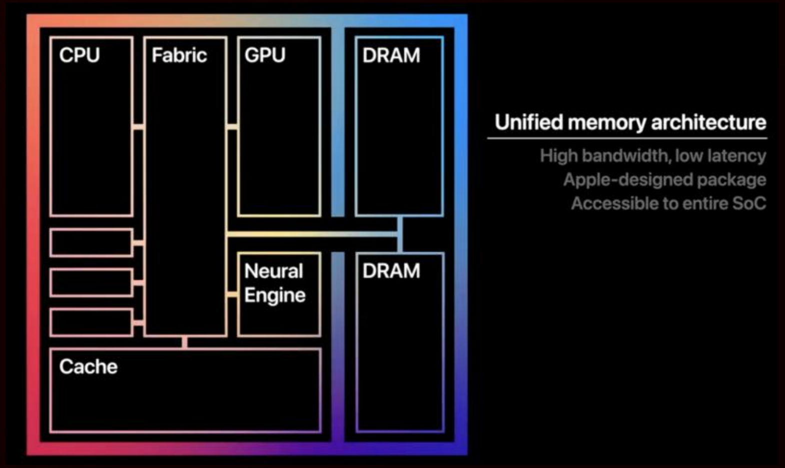
介绍


AMD
RYZEN AI
MAX Series
LPDDR5X
Ryzen AI Max+ 395


LPDDR5X
LPDDR5X
LPDDR5X
LPDDR5X
LPDDR5X
LPDDR5X
LPDDR5X
LPDDR5X
TOP
GZ302EA MAIN BO

隔壁棚的设计





GPU可以直接访问内存

内存 = 显存，显存 = 内存

LLM 本地推理模拟跑分

本地小模型

LM Studio

并非实际数据，仅作为示例展示！

LLM 本地推理模拟跑分

本地大模型

LM Studio

并非实际数据，仅作为示例展示！

跑分

电子斗蛐蛐

3D Mark Timespy

GPU在DX12 图形API分辨率为2K下的理论跑分

Cinebench R23

Multi Core多核分数

CINEBENCH

*N3B为TSMC第一代3nm技术节点，N3E为TSMC第二代3nm技术节点，N3P为TSMC第三代3nm技术节点。
*Ryzen AI Max+ 395为基准。

*分数越高越好。

Cinebench R23

Single Core单核分数

CINEBENCH

*N3B为TSMC第一代3nm技术节点，N3E为TSMC第二代3nm技术节点，N3P为TSMC第三代3nm技术节点。
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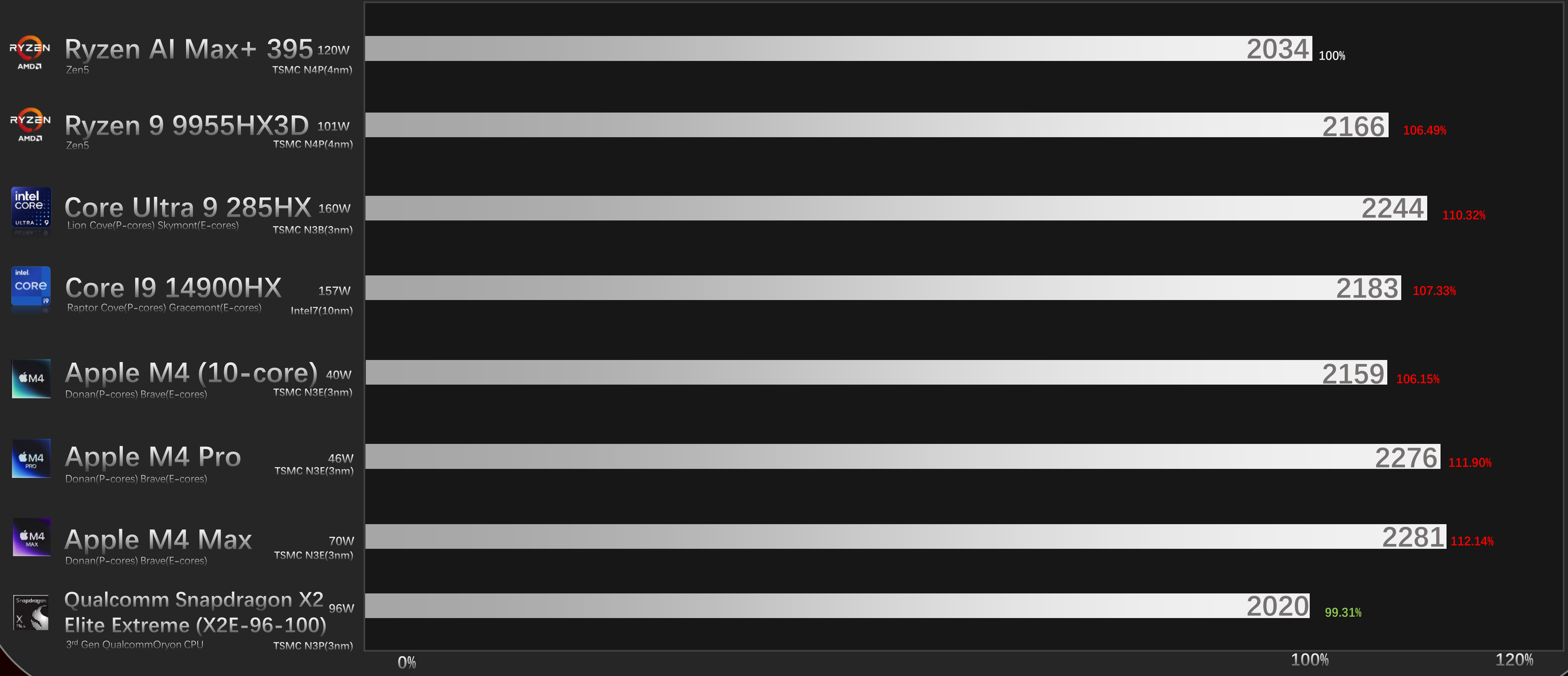
*分数越高越好。



Cinebench R23

Single Core单核分数

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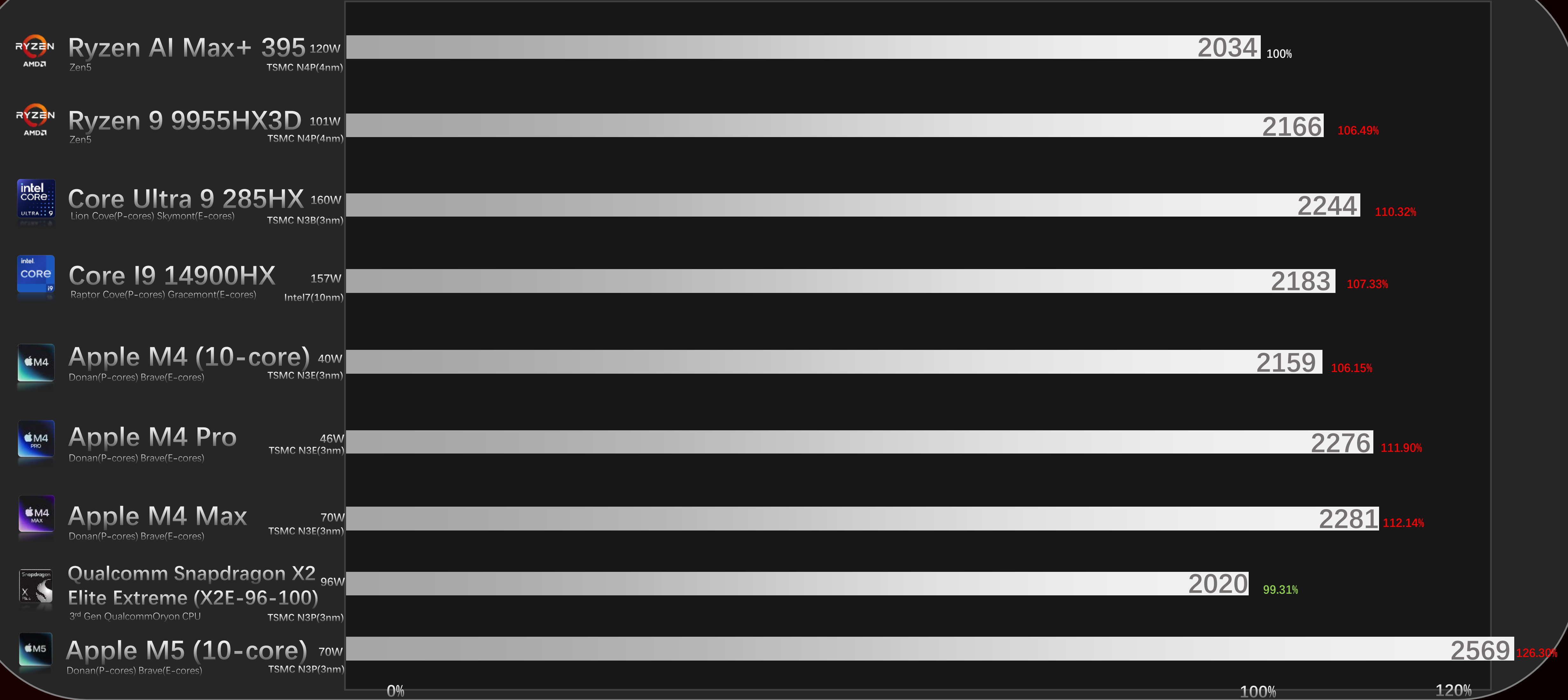


总结

16x
Zen5 CPU

TSMC
N4P

Unified Memory
Architecture

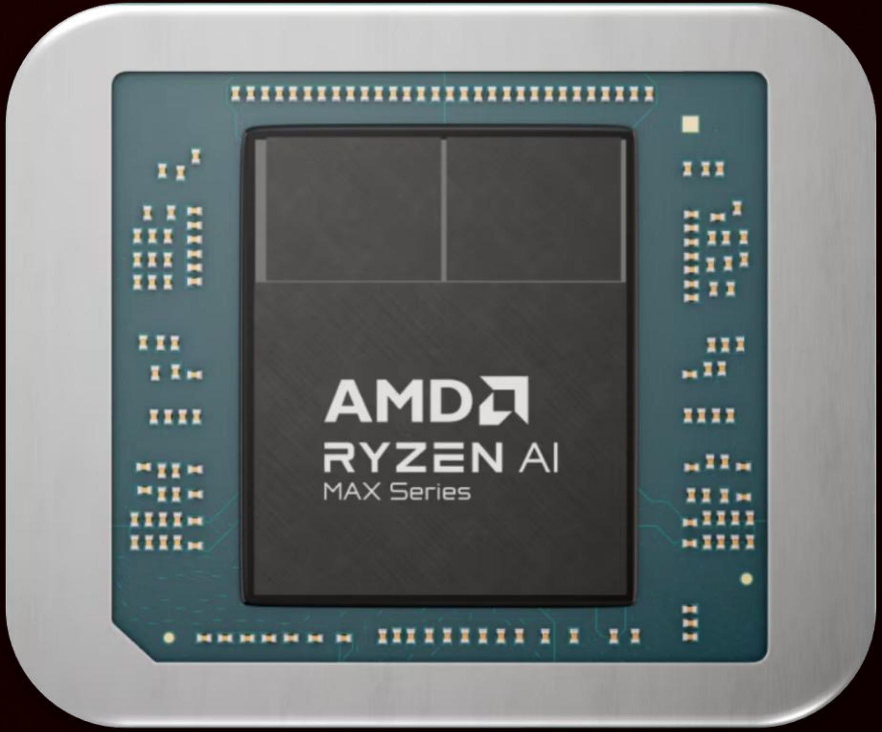
50 TOPS
NPU

LPDDR5X-8000
Up to 128GB

Radeon 8060S
RDNA3.5

40
CUs

2560
Shading unit



126 TOPS
Overall

PBO

Ryzen
AI

PCIE
4.0

configurable
TDP

Start from *only* NT$69999 ~ NT$84999